P2P PROCESS ANALYSIS - SQL Server DOCUMENTATION

Author: Aleksandra Laskowska
www.linkedin.com/laskowskaaleksandra

May 2026

Contents

1. Project Overview

This project presents analysis of the Purchase-to-Pay process using event log data. The aim was to evaluate recent P2P process performance using an as-of-date approach. The analysis focused on three business areas:

- SLA performance over time and backlog risk
- Identification of process bottlenecks across process stages
- Identification of segment-level risk concentration drivers

The project was built as a practical SQL analysis of a realistic business case.

2. Dataset & Process Context

2.1 Data Model

The project uses a dataset based on BPI Challenge 2019 – Purchase Order Handling. The data was transformed into a relational model using fact and dimension tables. Core entities include:

- fact_cases
- fact_events
- dim_activity
- dim_vendor
- dim_company
- dim_spend_area

The process model was simplified to four main process milestones presented in process order:

- Create Purchase Order (PO),
- Record Goods Receipt (GR),
- Record Invoice Receipt (IR),
- Clear Invoice (CI).

Data preparation was completed with AI assistance based on defined process rules and controlled validation of results (more details in chapter 3.1).

2.2 Process Definitions

The event log is built using cases and events. A case represents a single process instance related to purchasing goods or services. Events represent recorded updates during process execution.

Lead Time was defined as the duration between PO and CI milestones. Only closed cases with both milestones recorded were included in SLA analysis.

To define the SLA threshold, lead time distribution and historical trends were reviewed. Based on this analysis, 75 days was selected as the SLA limit.

Cases without a Clear Invoice milestone were classified as open cases. In the SQL logic these cases were analysed using Age Days.

2.3 As-of-Date Approach

The source dataset contains historical event log data. To represent a more realistic business scenario, the analysis was built using an as-of-date approach.

After the last day with new PO creation, the volume of new events in the dataset decreased significantly. Based on this behaviour, the last active PO date was selected as the as-of-date reference point. This date was define point for open ageing and rolling-window analysis.

<graph present trend new PO for new cases vs time>

2.4 Timestamp Logic

To calculate process stage duration consistently, the first recorded event was used for the Create Purchase Order milestone. To extract the first recorder this timestamp for every cases, in SQL logic this was implemented using the MIN function.

For the remaining milestones, the last recorded event was used to represent final stage completion. In SQL logic this was implemented using the MAX function.

3. Data Preparation & Validation

3.1 Data Preparation

Data extraction from the original event log was prepared with AI assistance. I defined the required output structure, target tables, and process milestones, and then validated the generated results against expected process logic.

The generated code was reviewed through output checks and used to prepare staging CSV files for SQL analysis.

3.2 Analytical Base Preparation

As the first step of analysis, case-level datasets were prepared for each business question. Each table contained information at single case level.

From a technical perspective, the required logic was prepared using CTEs and then loaded into reusable #temp tables. This approach reduced repetition of the same SQL logic across different analyses.

Depending on the analysis topic, the analytical base included information such as Start Time, End Time, classification cases information (Lead Time vs Age Days, Delay Flag or risk buckets), and stage duration.

For segment-level analysis (vendor, company, spend area), a common base table was first prepared with all required case information and segment keys. This table was then used to prepare dedicated analytical tables for each segment type.

3.3 Data Validation

During preparation of the first analytical base table, event log quality was validated. The validation included checks for missing milestones and incomplete process paths. It was verified whether closed cases contained all required milestones between process start and process end.

Stage duration calculations were also validated to identify negative duration values. These cases were most likely caused by inconsistencies in milestone update logic or event record order.

The validation showed that cases with negative duration represented below 5% of the total population. Due to the relatively low share and lack of additional source information explaining these inconsistencies, these cases were excluded from further duration analysis.

To better understand invalid and incomplete process paths, Primary Issue logic was implemented during validation. Primary Issue represented the earliest issue preventing a case from becoming a valid and complete process flow. Two main issue types were identified:

- Missing milestone – when one of the required process milestones was not recorded,
- Invalid timestamp – when the timestamp sequence created negative stage duration.

For missing milestones, the first unavailable stage was treated as the Primary Issue. For invalid timestamps, the issue was assigned to the first stage with negative duration.

This classification was later used to understand which case populations could not be included in duration and bottleneck analysis.

Validation rules defined during the first analysis stage were later reused across the remaining SQL analysis. Validation was performed iteratively during SQL development. Intermediate tables and aggregated outputs were reviewed multiple times to verify consistency of logic and calculated metrics.

For segment analysis, Impact Share validation was also performed. The total ratio of delayed cases across all segments was checked against the total delayed case population. Small differences below 1% were accepted as rounding differences.

Additional validation was performed to ensure that each segment contained enough cases to represent a meaningful sample. A minimum threshold of 100 cases per segment was applied to reduce analytical noise.

3.4 Filtering & Population Rules

To analyse SLA performance, two main case categories were defined:

- Closed cases (Lead Time in SQL logic) – cases with recorded CI milestone,
- Open cases (Age Days in SQL logic) – cases without recorded CI milestone.

As described in chapter 3.3, cases with negative duration values were excluded from duration analysis.

Stage duration was calculated only for cases with all required timestamps available for a given process stage. As a result, some open cases could still be included in partial stage duration analysis if the required milestone sequence was available.

Bottleneck contribution was calculated only for closed, valid, and complete cases. This was required to compare stage contribution against the same total Lead Time across all analysed cases. Open cases were excluded from this part of analysis.

Additional filtering rules were applied for segment analysis. Segment ranking was based on closed cases with validated timestamps and a minimum threshold of 100 cases per segment. Further detailed analysis focused only on the top 10 or top 15 delay drivers based on Impact Share values.

4. Analytical Approach

4.1 SLA & Backlog Analysis

As the first step of SLA analysis, the distribution of Open Cases vs Closed Cases was reviewed to understand the latest process status. Closed Cases represent final performance, while Open Cases represent current operational risk.

The next step included monthly trend analysis of Closed vs Open Cases. This approach helped define the normal distribution of SLA performance and identify periods with increased risk of lower SLA performance. Additional drill-down analysis was prepared for Closed Cases to review the share of cases closed within the SLA window.

Monthly analysis was used to observe longer-term performance trends, while weekly analysis focused on recent operational risk for the current month and previous three months.

To better identify risk level, a monitoring point was defined using the median duration of SLA-compliant closed cases. Open Cases between this monitoring point and the SLA threshold should be prioritised for monitoring and potential intervention due to higher risk of exceeding the SLA window.

To better understand backlog behaviour (Open Cases over 90 days), bucket logic was implemented to classify cases based on duration ranges. This helped determine whether backlog growth was a long-term issue or mainly related to recent periods, where results may still be partially affected by data immaturity.

The final analytical flow separates final SLA performance from current operational risk. This approach provides a more complete process overview than analysing only closed cases.

4.2 Bottleneck Analysis

Bottleneck analysis was performed only on closed cases with validated stage duration. Before final analysis, the distribution of missing milestones and invalid timestamps was reviewed to understand which populations were excluded. Before final bottleneck analysis, the distribution of invalid and incomplete process paths was reviewed using Primary Issue classification.

To identify bottlenecks correctly, stage duration was analysed first. Median and P90 values were used instead of averages due to skewed duration distribution and outliers. The analysis compared SLA-compliant and delayed cases. Larger differences between these two populations indicated higher probability that a stage contributed to process delays.

To confirm the bottleneck stage, Bottleneck Contribution was calculated as the ratio of stage duration to total case Lead Time. This approach helped distinguish stages with long duration from stages with the highest impact on total process performance.

4.3 Segment Analysis

Each segment type – Vendor, Spend Area, and Company – was analysed separately using the same analytical approach. The first step was preparation of separate rankings based on Impact Share, Delayed Rate, and Delayed Volume.

Impact Share was used to identify segments with the highest business impact based on delayed case population. Delayed Rate represented the share of delayed cases within the population of a single segment individual.

To focus on the most significant drivers, deeper analysis was limited to top contributors selected using Pareto logic based on Impact Share values.

Detailed analysis included stage duration and Bottleneck Contribution, following the same approach as described in section 4.2. The analysis focused mainly on identified bottleneck stages.

Segment-level results were compared against SLA and delayed populations, as well as against the overall SLA population baseline. This comparison helped identify whether delays were related to the same bottleneck stage or caused by different process behaviour.

Spend Area analysis additionally included category and sub-category review to identify which business areas generated the highest delay concentration.

For Company analysis, drill-down bottleneck analysis was not expanded further because one company represented almost the entire delayed population. In this situation, additional bottleneck comparison would mainly duplicate existing findings.

5. Key Findings

- SLA performance for closed cases remained relatively stable during the analysed period. However, the share of open cases increased in recent periods, which may indicate growing execution backlog.
- Open case analysis showed increased risk concentration between the internal monitoring point and SLA threshold window.
- The main bottleneck was identified between Invoice Receipt and Clear Invoice (IR → CI). This stage had the highest contribution to total lead time in delayed cases.
- Delay concentration was identified across selected vendors and spend areas. Top vendors and top spend areas represented the majority of delayed case volume.
- Packaging was the most critical spend area category. Within this category, Labels represented the highest delay concentration.

- One company entity represented almost the entire delayed population, which may suggest local process or governance issues.

6. Limitations

- Analysis limited to the available event-log dataset.
- Dataset does not allow direct identification of operational root causes of delays.
- No process ownership or organisational structure data available.
- Vendor and company names anonymised in the source dataset.
- Data represents a historical process snapshot rather than live operations.
- Results for the most recent periods may be partially affected by incomplete process closure.